

Physics Internal Assessment: Fructose Solution Concentration Determination

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2371 words

1 Introduction

The determination of sugar concentration in a solution is key for many industries including food processing, medical testing and chemical production, as it is directly linked with its properties and quality stability. An example of this is that foods high in sugar are more resistant to storage, and less prone to microbial contamination (Gong). Traditional measurement methods of sugar concentration within a solution include colorimetry and optical low-coherence interferometry, which require very “strict control of experimental conditions and are susceptible to interference from environmental factors.” (Gong). However, this research aims to delve into the area of polarization within the field of optics, as sugar is able to rotate the plane of polarization of polarized light. This then led to my research question: How does the mass of sugar (fructose) dissolved in water affect the angle of polarization of polarized light, when the temperature and solution depth are kept constant?” To answer this question, I developed a mathematical relationship through research, I conducted an experiment, analyzed the results and compared it to my initial hypothesis.

2 Background

Light is an electromagnetic wave that could be polarized using a polarization filter. When light becomes polarized, it is specifically the electric field component that is restricted to oscillate in a single plane, while the magnetic field follows accordingly. The polarized light would then only oscillate in a single plane. Certain materials rotate the angle of polarization of light when it shines through, called optical activity. Dextrorotatory materials are optical active materials that rotate light clockwise, while levorotatory anticlockwise (Elert).

Sugar, such as glucose and fructose, are classic examples of optically active substances. While glucose rotates polarized light clockwise, fructose rotates it counterclockwise due to

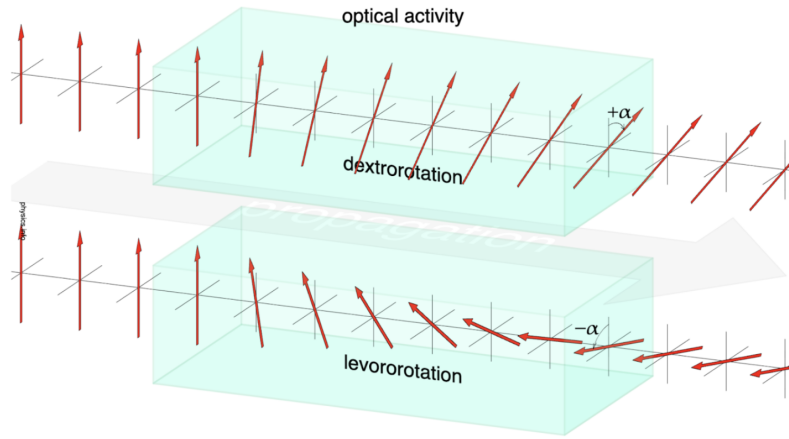


Figure 1: Diagram showing the difference between dextrorotation and levorotation (Elert)

its molecular structure. The angle of polarization depends on several factors, including the concentration of the solution, the path length of light through the medium, the temperature, and the wavelength of the incident light (Elert).

According to Biot's law, the angle α , at which the plane of polarization is rotated, depends on the mass concentration C_m of the sugar solution, on the path length l , and on a material-specific quantity called the specific rotation s . This is expressed as:

$$\alpha = s l C_m$$

(Elert)

This relationship forms the theoretical basis for predicting how the mass of fructose dissolved in water affects the angle of polarization.

3 Hypothesis

Based on the theoretical model above, the angles of polarization should have a proportional relationship with the mass concentration of the sugar solution, since both l and C_m are kept constant. A sketch of my hypothesis is shown below:

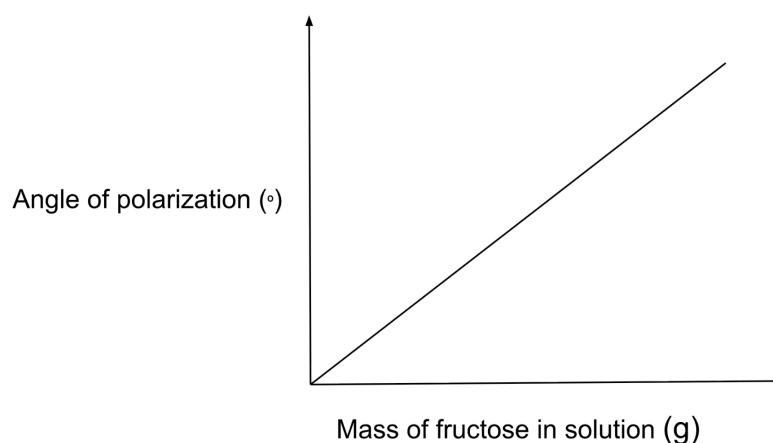


Figure 2: Diagram showing the hypothesis of the relationship between angle of polarization and mass of fructose in a solution

4 Methodology

For my experiment, I had a beaker containing the solution. The beaker was 5.00cm (± 0.01 cm) wide, and was placed in between two polarizing filters. I had one white LED that provided a narrow beam going through the solution. I looked at the color of the resulting light beam, and rotated the second polarizing filter to determine the angles of polarization. The final angle of polarization is then measured using a protractor.

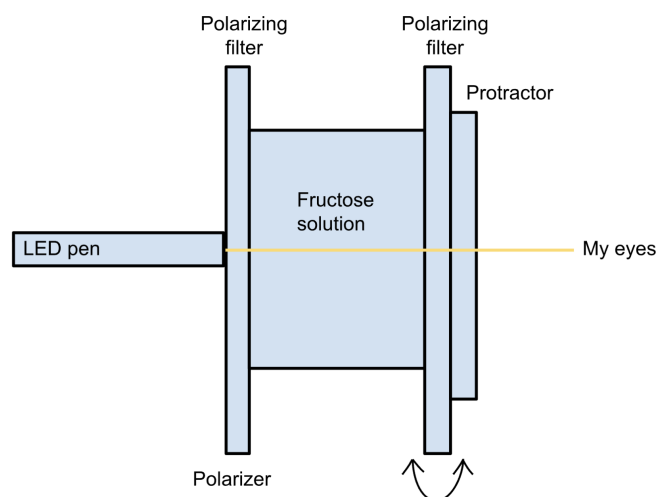


Figure 3: Diagram of the side view of the experiment setup

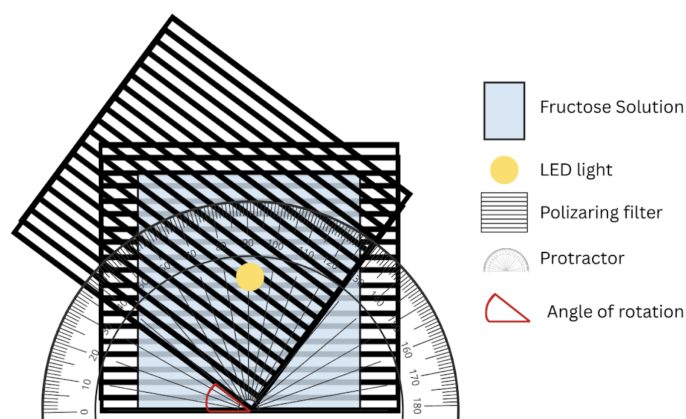


Figure 4: Diagram of the front view of the experiment setup

The independent variable was the mass of fructose dissolved in 30.00 ± 0.01 g of tap water. I added 6.00 ± 0.01 g of fructose each time, reaching up to 72.00 g in total. Both the water and the fructose were measured using a digital balance with an uncertainty of ± 0.01 g. Fructose was chosen because it can theoretically dissolve up to 400% in water at room temperature (approximately 28°C), providing a large possible range for the independent variable. However, as I approached 72.00 g of fructose in 30.00 g of water, the dissolution time increased significantly, reaching over 30 minutes, and the solution became increasingly slow to return to transparency. To balance practicality and accuracy, I stopped at the ratio of 72.00 g to 30.00 mL.

To calculate the mass concentration of fructose in the solution, I used:

$$\text{Concentration} = \frac{\text{mass of fructose}}{\text{mass of water}}.$$

Since the mass of water was kept constant at 30.00 ± 0.01 g, the concentration increased linearly as 6.00 g increments of fructose were added.

The dependent variable was the angle of polarization rotation of the transmitted light. To measure this, I observed the color of the resulting light beam while rotating the second polarization filter, as shown in the previous figure. When the beam appeared darkest, I recorded the corresponding angle using a protractor with an uncertainty of $\pm 1^\circ$. A protractor was used due to accessibility, although a digital polarimeter would have offered superior precision. Preliminary testing showed that after three trials, the results were consistent and the experimental uncertainty was below 5%, so each measurement was repeated three times.

Several control variables were implemented to maintain accuracy. The depth of the solution (the optical path length) was kept constant at 5.0 ± 0.1 cm, determined by the diameter of the same beaker used throughout the experiment. The experiment was conducted over two sessions due to time constraints; this caused slight variations in room

temperature, which otherwise remained approximately constant at 26°C to 28°C with an uncertainty of $\pm 1^\circ\text{C}$. The same apparatus were reused throughout, and the digital balance maintained an uncertainty of $\pm 0.01\text{ g}$.

The procedure was efficient. For each new concentration, I added 6.00 g of fructose to the existing solution and then re-set the polarizers, light source, and protractor, as this step was the main contributor to random error. This allowed each set of three trials to be recorded within about five minutes. However, as the concentration increased, the dissolution time rose dramatically, reaching 15–20 minutes of continuous stirring at the highest concentration (72.00 g/30.00 mL). I avoided heating the solution to maintain temperature stability throughout the experiment.

There were no significant safety risks involved. The white LED used had low intensity and did not pose a danger to the eyes. The fructose solution was safe to handle, and all materials were cleaned appropriately after the experiment. No biological or ethical concerns were associated with this investigation.

5 Data

Table 1: Data table showing the relationship between the amount of fructose in the solution and the angle of polarization.

fructose / g		angle of rotation / $\pm 1^\circ$			angle of rotation / $^\circ$		
mass	Δ	Trial 1	Trial 2	Trial 3	Average	Δ	Percentage Δ / %
0.00	0.00	0	0	0	0	0	0.00
6.00	0.01	-7	-6	-6	-6	1	0.91
12.00	0.02	-12	-10	-12	-11	1	1.0
18.00	0.03	-20	-18	-20	-19	1	0.60
24.00	0.04	-24	-22	-23	-23	1	0.43
30.00	0.05	-27	-26	-28	-27	1	0.37
36.00	0.06	-29	-28	-31	-29	2	0.52
42.00	0.07	-32	-33	-34	-33	1	0.30
48.00	0.08	-35	-37	-36	-36	1	0.28
54.00	0.09	-38	-39	-39	-39	1	0.15
60.00	0.10	-43	-42	-42	-42	1	0.14
66.00	0.11	-46	-46	-45	-46	1	0.13
72.00	0.12	-52	-51	-53	-52	1	0.19

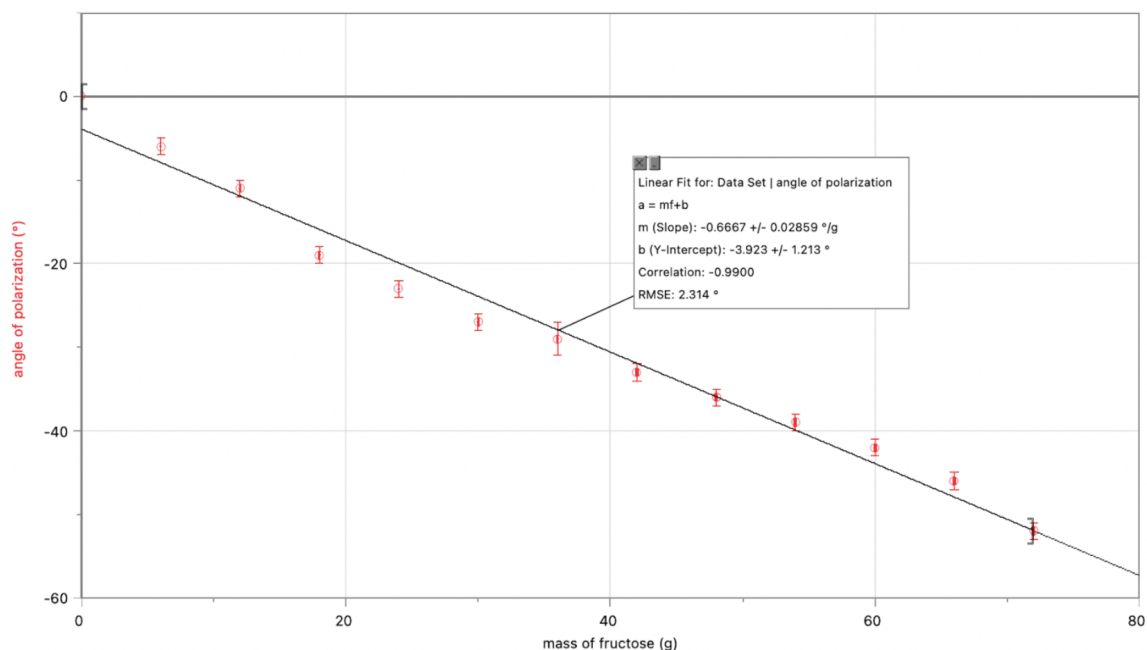


Figure 5: Figure to show the relationship between mass of fructose in the solution and the angle of polarization

6 Analysis

My data shows a negative linear relationship between the mass concentration of fructose dissolved in 30.00 mL of water and the rotation angle of polarization. As the mass of sugar increases from 0.00 g to 72.00 g, the rotational polarization angle also increases from 0° to 52° linearly. The equation suggested by my results indicates that

$$\alpha = mf + b$$

where the gradient is $m = 0.67 \pm 0.03$ degrees/g and the intercept is 4 ± 1 degrees.

The pattern within the data is quite clear and there are no extreme anomalies. Although there are indeed ups and downs in data points that are not exactly on the line of best fit, which do not go through the error bars either, these slight deviations are not significant and consistent enough to support a completely different fit for the graph.

There are two methods of approaching the uncertainty of the line of best fit. One would be the one determined by Logger Pro, while the other would be drawing the max and min lines through the uncertainty bars. It can be noticed that my line of best fit does not necessarily go through all of the error bars. On the contrary, it goes through the minority of them. Hence it would be very inaccurate if I determined the uncertainty of my line of best fit through drawing max and min lines manually. Thus, it was decided to rely on Logger Pro's uncertainty calculation.

The percentage uncertainty for the dependent variable is higher at lower angles, due

to the limitation of the measurement tool. For example, the $\pm 1^\circ$ at 6.00 g of sugar meant a 16% uncertainty, because the angle of polarization rotation was only 6° ; while at higher angles the $\pm 1^\circ$ was much less significant. For example, for the measurement of 72.00 g of fructose, it meant a 2% uncertainty because the angle of rotation was 52° . This explains why the error bars became smaller as the independent variable increased. Overall, the resulting correlation is -0.99 , which indicates a relatively high correlation. The RMSE value of 2.314 also indicated that on average, my data points were only about 2.3° away from the line of best fit, which is an uncertainty of about 4%, since my dependent variable ranged from 0 – 52° . This again suggests a very high correlation for a linear trend.

Comparing the experimental results to the research, fructose has a specific rotation value of $-92^\circ \text{ mL g}^{-1} \text{ dm}^{-1}$ (Lide) (IO4 - Polarisation and Saccharimetry.), which, when substituting the experimental condition, yields the following slope:

$$-92^\circ \cdot 30 \text{ mL} \cdot 0.5 \text{ dm} = -1.533^\circ/\text{g}.$$

Thus, theory predicts that the fructose solution, under my settings, should have a slope of $-1.53^\circ/\text{g}$. While my experimental value shows a $-0.67^\circ/\text{g}$ relationship, the fact that the sign aligns shows that fructose is indeed levorotatory, meaning that it rotates the plane of polarization anti-clockwise. However, when we compare the actual results, we notice that the measured slope is about 43% of the expected theoretical slope.

Overall, this result partially supports my hypothesis. The data clearly shows a proportional and linear relationship between the mass concentration of fructose and the rotation angle, which matches the mathematical model. However, the difference in magnitude between the theoretical and experimental slopes suggests that although the general trend is correct, the numerical agreement is limited by experimental and methodological factors.

7 Evaluation

Systematic Errors

There were a few areas where systematic errors could have been present in experiment.

First of all, because I decided to use a white light pen instead of a monochromatic light source, there was no real “point of extinction”, where all light going through is blocked. This is because a white LED contains a wide range of wavelengths, each of which will be interfered differently. Hence, I observed a gradual transition from yellow to blue to purple as the polarizing filter was rotated. As a result, I had to subjectively determine the “darkest” color. I chose a rather dark blue and tried to maintain consistency by looking for the same color throughout the experiment and across all trials. This may be the source of a significant systematic error. Because I chose blue instead of purple, my wavelength reduces and hence greatly reduces my observed angle of rotation compared to the actual angle of rotation. This results in a deduced slope value. This marks a

methodological weakness which could be improved by using a monochromatic light source such as a sodium lamp or a laser, which could have provided me with a more clear angle of distinction.

Another systematic error that I have noticed was that at higher concentration levels, the sugar solution struggled to dissolve and to return to transparent. This could have slightly lowered the effective concentration levels, hence reducing my observed slope. However, the fact that my data supports a linear trend suggests that this was a minor methodological limitation. By using a magnetic stirrer instead of my doing it by hand, could have allowed the solution to be better fully dissolved. Moreover, allowing a fixed time for the solution to turn transparent would also have helped reducing some scatter.

Random Errors

Alongside systematic errors, there are also random errors involved in my experiment.

Firstly, how I subjectively judged the "darkest color" is a source of random error. Since it is done by my human eye, and it was done within a short period of time, I may have subconsciously skewed the results. Moreover, on a larger time scale, which is the time in between each change in the independent variable, I might forget the specific shade of blue that I rely on. This is a methodological limitation. A possible improvement that would reduce this random error is to have conducted the experiment in a darker environment, with fixed positions of the tools and the viewer. This could have increased the consistency of observations.

Another significant source of random error comes from the precision of the protractor used to measure the angle of polarization. The protractor had an uncertainty of $\pm 1^\circ$, which is relatively large when thinking of the measured angles of rotation at lower fructose concentrations. This directly influences my outcome. An improvement that could have significantly increased the precision would have been to use a digital polarimeter. For example, a Schmidt + Haensch UniPol V Digital Polarimeter has a precision of $\pm 0.005^\circ$, which is 200 times more precise than the protractor I used. Using this would have greatly reduced my random error and made my values more accurate.

Extensions

There are several possible extensions of this investigation. One would be to repeat the entire experiment using monochromatic light to determine how closely the measured slope can match the theoretical prediction. Another extension would be to study how the rotation depends on wavelength by measuring the angle for different LED colours, which would explore optical rotatory dispersion. I could also compare fructose with glucose or sucrose to examine how different molecular structures influence rotation. Finally, investigating the effect of temperature on the angle of rotation would provide insight into another variable not fully controlled in this experiment.

8 Conclusion

My research question was: *How does the mass of fructose dissolved in water affect the angle of polarization of polarized light, when the temperature and solution depth are kept constant?*

My experimental results show a clear negative linear relationship between the mass concentration of fructose and the rotation angle of polarization. The relationship matches the theoretical model in terms of direction, since fructose is expected to be levorotatory, which my data confirms. The magnitude of the measured slope, however, is only about 43% of the theoretical prediction. Therefore, while the overall behaviour supports Biot's law and supports my hypothesis of a proportional relationship, the specific rotational angle does not fully match the theoretical value.

The limitations of the experimental setup, including the use of white light, subjective colour judgement, and the precision of the protractor, reduce the accuracy of my results. Thus, the research question is answered, but with notable limitations regarding the precision and accuracy of the measured angle of rotation.

Overall, the experiment successfully identified the expected linear trend. However, the limitations of the method, particularly the use of white light and low-precision angle measurement, restrict how precisely the research question can be answered. The general relationship is clear, but the magnitude cannot be determined with high accuracy under the conditions of this investigation. A more controlled optical setup would be required to reach a fully reliable numerical comparison with theory.

References

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